Notes: Bajari, Hong, and Ryan (Econometrica, 2010)

Identification and Estimation of a Discrete Game of Complete Information

Contents

1 Big picture

- **Research question.** How can researchers identify and estimate static discrete games of complete information when the game can have multiple Nash equilibria, including mixed strategy equilibria?
- **Core econometric problem.** In a standard discrete choice model, primitives map to choice probabilities. In a complete-information game, a given payoff vector may generate many equilibria, so the primitives do not determine a unique outcome unless the econometrician specifies how equilibria are selected.
- **Main idea.** Treat equilibrium selection as part of the empirical model. The primitives are not only player utilities, but also an equilibrium selection mechanism, λ, which assigns probabilities to equilibria in the equilibrium set.
- **Key methodological contribution.** The paper proposes simulation-based estimators for general finite normal-form games of complete information. The estimator computes all Nash equilibria, allows mixed strategies, and uses importance sampling to reduce the computational burden.
- **Identification contribution.** The paper gives two identification strategies: identification at infinity and exclusion restrictions based on agent-specific payoff shifters.
- **Application.** The empirical application studies entry into California highway procurement auctions. Four large construction firms decide whether to incur bid-preparation costs and enter first-price sealed-bid auctions.
- **Main empirical finding.** In the auction-entry application, equilibria are more likely to be selected when they maximize joint profits, have a high Nash product, are not Pareto dominated, and, according to the point estimate, involve mixed rather than pure strategies.
- **Economic takeaway.** Multiple equilibria are not just a nuisance to be ignored. Equilibrium selection can be estimated, can be economically informative, and is necessary for simulation and counterfactuals in complete-information games.

2 Incremental contribution of the paper

- **General normal-form games.** Earlier empirical work often focused on special entry games or games with monotone structure. This paper develops a framework that can be applied to general finite normal-form games of complete information.
- **Mixed strategy equilibria.** The paper explicitly allows mixed strategies. This matters because, under broad payoff support, some payoff realizations produce no pure strategy equilibrium.
- **Estimated equilibrium selection.** Rather than assuming random selection, selecting an extremal equilibrium, or using bounds without specifying selection, the paper models and estimates the probability that each equilibrium is played.
- **Simulation and counterfactuals.** Because λ is estimated, the researcher can simulate outcomes under the model. This is difficult in approaches that identify only payoff inequalities or bounds.

- **Computational innovation.** The estimator uses a smooth importance sampler that avoids recomputing the equilibrium set each time the structural parameters change.
- **Identification theory.** The paper shows why unrestricted nonparametric identification generally fails, then gives sufficient conditions under which payoff functions and selection probabilities can be recovered.
- **First empirical test of selection criteria.** In the auction application, the paper estimates whether equilibria that are efficient, undominated, pure or mixed, and Nash-product dominant are more likely to be observed.

3 Model

3.1 Complete-information normal-form game

There are N players. Player i has a finite action set A_i. Let

$$A = \times_i A_i, \qquad a = (a_1, \dots, a_N) \in A.$$

The utility of player i from action profile a is

$$u_i(a, x, \theta_1, \epsilon_i) = f_i(x, a; \theta_1) + \epsilon_i(a),$$

where f_i is the deterministic payoff component and $\epsilon_i(a)$ is a payoff shock observed by the players but not by the econometrician.

The complete-information assumption means that all players know the full payoff vector when choosing strategies. The econometrician observes covariates and outcomes, but not the payoff shocks.

3.2 Equilibrium set

For a given payoff vector u, let $E(u)$ denote the set of Nash equilibria. This set can include both pure and mixed strategy equilibria. A mixed strategy equilibrium is denoted

$$\pi = (\pi_1, \dots, \pi_N),$$

where $\pi_i(a_i)$ is the probability that player i takes action a_i.

The important feature is that $E(u)$ is often not a singleton. The model therefore needs an additional object to map payoffs into observed outcomes.

3.3 Equilibrium selection mechanism

The paper introduces an equilibrium selection mechanism

$$\lambda(\pi; E(u), \beta),$$

which is the probability that equilibrium $\pi \in E(u)$ is selected. It must satisfy

$$\sum_{\pi \in E(u)} \lambda(\pi; E(u), \beta) = 1.$$

A useful parametric form is a multinomial-logit selection rule:

$$\lambda(\pi; E(u), \beta) = \frac{\exp\{\beta' y(\pi, u)\}}{\sum_{\pi' \in E(u)} \exp\{\beta' y(\pi', u)\}},$$

where $y(\pi, u)$ can include equilibrium characteristics such as:

- whether the equilibrium is pure;
- whether it is Pareto dominated;
- whether it maximizes the sum of player utilities;
- whether it has a high Nash product.

Interpretation. Selection is not an auxiliary computational trick. It is an economic object. It tells the researcher which equilibrium features are empirically associated with observed play.

3.4 Probability of an observed action profile

The probability of observing action profile a is

$$P(a \mid x, f, \lambda) = \int \sum_{\pi \in E(u(f,\epsilon),x)} \lambda(\pi; E(u(f, \epsilon), x)) \left[\prod_{i=1}^{N} \pi_i(a_i)\right] g(\epsilon) d\epsilon.$$

This expression has three steps:

- draw unobserved payoffs;
- compute the full set of Nash equilibria for those payoffs;
- average observed action probabilities over equilibria using the selection probabilities.

3.5 Why mixed strategies are necessary

The paper uses the matching-pennies logic to show that pure strategies alone are not enough. Matching pennies has only a mixed strategy equilibrium. Small perturbations of payoffs around matching pennies still produce games with no pure equilibrium. Therefore, if payoff shocks have sufficiently rich support, a complete-information discrete game will generate mixed-only equilibrium realizations with positive probability.

Restricting the estimator to pure strategies would then make the model ill-defined for some payoff draws. The paper's estimator avoids this problem by computing both pure and mixed equilibria.

3.6 Comparison with incomplete-information games

Incomplete-information games are often easier to estimate because they can produce smooth choice probabilities and sometimes allow two-step estimation. But the equilibrium structure is different. Private information can effectively refine the equilibrium set. Complete-information games, by contrast, can have many equilibria, and the number of equilibria tends to grow with the number of players and actions.

The paper does not claim that complete information is always better. The point is that if the application is naturally complete information, the econometric model must handle multiplicity directly.

4 Simulation and estimation

4.1 Naive simulator and computational problem

A direct simulator would repeatedly draw payoff shocks ϵ, compute utilities, solve for the full equilibrium set $E(u)$, and average predicted actions over equilibria. This is conceptually simple but computationally expensive because the equilibrium set must be recomputed for each parameter vector during estimation.

The computational burden can be severe. The paper reports that solving for all equilibria in a five-player, two-action game can take up to 20 minutes on a single processor workstation. Repeating that inside an optimizer is infeasible for many applications.

4.2 Importance-sampling idea

The paper changes the integration variable from payoff shocks ϵ to utility indexes u. Let $h(u \mid \theta, x)$ be the density of utility vectors implied by the structural parameters. Then

$$P(a \mid x, \theta, \beta) = \int \sum_{\pi \in E(u)} \lambda(\pi; E(u), \beta) \left[\prod_{i=1}^{N} \pi_i(a_i) \right] h(u \mid \theta, x) du.$$

Using an importance density $q(u \mid x)$, the simulated probability is

$$\widehat{P}(a \mid x, \theta, \beta) = \frac{1}{S} \sum_{s=1}^{S} \sum_{\pi \in E(u^{(s)})} \lambda(\pi; E(u^{(s)}), \beta) \left[\prod_{i=1}^{N} \pi_i(a_i) \right] \frac{h(u^{(s)} \mid \theta, x)}{q(u^{(s)} \mid x)}.$$

The key trick is that the draws $u^{(s)}$ are drawn from q, not from the parameter-dependent structural density. Thus the equilibrium set $E(u^{(s)})$ can be computed once and reused as θ and β change.

4.3 Why the simulator is useful

The simulator has three important properties:

- It is unbiased for $P(a \mid x, \theta, \beta)$ for a valid importance density.
- The equilibrium sets are independent of the current parameter values after the utility draws are fixed.
- The simulated probability is smooth in the parameters, which makes numerical optimization more stable.

The importance density must have sufficiently thick tails so that the likelihood ratio h/q does not become unstable. In the application, starting values and the importance density are constructed from a simpler private-information version of the model.

4.4 MSM estimator

The data are a sequence (a_t, x_t) for $t = 1, \ldots, T$. For each action profile $k \in A$, the true parameters imply

$$E\left[1(a_t = k) - P(k \mid x_t, \theta, \beta)\right] w_k(x_t) = 0,$$

where $w_k(x_t)$ is a vector of instruments or weight functions.

The sample moments are

$$m_{k,T}(\theta, \beta) = \frac{1}{T} \sum_{t=1}^{T} \left[1(a_t = k) - \widehat{P}(k \mid x_t, \theta, \beta)\right] w_k(x_t).$$

For a positive definite weighting matrix W_T, the estimator is

$$(\widehat{\theta}, \widehat{\beta}) = \arg\min_{\theta, \beta} m_T(\theta, \beta)' W_T m_T(\theta, \beta).$$

The paper emphasizes MSM rather than maximum simulated likelihood because MSM can be unbiased and consistent for a fixed number of simulations, whereas MSL generally requires the number of simulations to grow with sample size.

5 Identification and interpretation

5.1 Necessary normalizations

The paper imposes two standard restrictions.

Location normalization. For each player, the payoff of one action is fixed at a known constant, usually zero. Adding the same constant to all payoffs does not change best responses, so location is not identified without this normalization.

Scale normalization. Multiplying all payoffs by a positive constant does not change equilibria. The scale is pinned down by assuming a known distribution for payoff shocks, such as independent standard normal shocks.

5.2 Why unrestricted nonparametric identification fails

Holding x fixed, the econometrician observes action probabilities $P(a \mid x)$. In a two-by-two game there are four action probabilities, but only three independent probabilities because they sum to one. Even if the selection mechanism were known, there are four non-normalized utility parameters to recover.

Thus the unrestricted payoff functions are underidentified. The problem becomes worse in larger games because the number of payoff parameters grows faster than the number of independent action probabilities.

This negative result justifies the paper's use of structure: linear-index payoffs, exclusion restrictions, and parametric or semiparametric selection rules.

5.3 Identification at infinity

The first identification strategy assumes that payoffs are linear indexes and covariates have sufficiently rich support. The idea is to send some covariates to extreme values so that some players choose particular actions with probability approaching one.

At such values, the remaining player's problem becomes approximately a single-agent discrete choice problem. This allows the researcher to identify payoff parameters. Once payoffs are identified, the selection probabilities can be recovered in regions where utility shocks have little influence on total utility and the set of equilibria is sufficiently small relative to the number of observable action probabilities.

Interpretation. Identification at infinity uses large covariate values to remove strategic uncertainty locally. The game becomes simple enough to recover payoffs, then the recovered payoff matrix is used to study equilibrium selection.

5.4 Exclusion restrictions

The second strategy uses agent-specific payoff shifters. The key assumption is that for each player i there is a covariate x_i that enters player i's payoff but not the payoffs of other players and not the equilibrium selection mechanism.

This kind of exclusion restriction is natural in many applications:

- in airline entry, route-network or hub variables can shift one airline's payoff;
- in retail entry, distance from a distribution center can shift one firm's cost;
- in technology adoption, employee rank, job function, or location can shift one user's adoption value.

The intuition is that agent-specific shifters move one part of the payoff matrix while leaving other parts fixed. Sufficient variation in these shifters creates enough independent action-probability variation to identify the model.

5.5 Limits of identification

Even after payoffs are known, a completely unrestricted selection mechanism may not be identified if the number of equilibria exceeds the number of independent outcome probabilities. Therefore, λ cannot be arbitrarily flexible in large games. The empirical application uses a parsimonious logit selection rule based on economically interpretable equilibrium characteristics.

6 Application: California highway procurement auctions

6.1 Setting

The application studies CalTrans highway paving procurement auctions from 1999 to 2000. Contractors choose whether to enter by preparing and submitting a bid. If they enter, they participate in a sealed-bid first-price auction, and the lowest qualified bid wins.

This setting is attractive because:

- bids have fixed public deadlines;
- communication between bidders would be illegal collusion;
- contracts are single projects with fixed scope and duration;
- a static entry model is more plausible than in long-lived market-entry applications.

6.2 Bidding game

The paper models bidding as an asymmetric first-price auction with independent private costs. Bidder i has private cost c_{it} and submits bid b_{it}. If the set of bidders is $N(t)$, bidder i's expected profit from bid b_{it} is

$$(b_{it} - c_{it}) \prod_{j \in N(t), j \neq i} \left[1 - G_j(b_{it} \mid x_t)\right].$$

Following Guerre, Perrigne, and Vuong, the first-order condition implies the cost inversion

$$c_{it} = b_{it} - \left[\sum_{j \in N(t), j \neq i} \frac{g_j(b_{it} \mid x_t)}{1 - G_j(b_{it} \mid x_t)} \right]^{-1}.$$

This allows the researcher to recover bidder markups and expected entry profits from observed bids and estimated bid distributions.

6.3 Entry game

The paper focuses on the four largest firms. Entry by smaller fringe firms is treated as exogenous because solving a complete-information game with hundreds of potential entrants would be computationally infeasible, and fringe firms are unlikely to discipline the largest firms at the margin.

Let $a_{it} = 1$ if large firm i enters project t. Conditional on the entry vector a, the participating set is $N(t \mid a)$, which includes observed fringe entrants and those of the four large firms that enter. The entry payoff is expected bidding profit minus the fixed bid-preparation cost θ_i.

The exclusion variation comes mainly from distance and firm fixed effects. Each large firm has a different distance to each project, and distance shifts that firm's cost of bidding and winning without directly shifting rivals' costs in the same way.

6.4 Data

The sample contains:

- 414 CalTrans paving contracts;
- 1,938 bids;
- 271 bidders;
- \$1.326 billion in total winning bids.

For each project and bidder, the data include the bid, CalTrans engineer's estimate, distance to the project, utilization or capacity, and a fringe-firm indicator.

6.5 Estimation steps

The application proceeds in two steps.

Step 1: Recover expected bidding profits. The authors estimate bid distributions and markups using a semiparametric version of the Guerre-Perrigne-Vuong approach. Bid functions depend on the engineer's estimate, distance, utilization, fringe status, firm fixed effects, and project fixed effects.

Step 2: Estimate entry costs and equilibrium selection. Taking the expected profits from Step 1 as given, they estimate bid-preparation costs θ_i and the equilibrium selection parameters. The selection rule depends on whether an equilibrium is pure, joint-profit maximizing, Pareto dominated, and high Nash-product.

6.6 Main empirical interpretation

The estimates imply economically sensible bid-preparation costs. The profit scale implies that one unit of estimated fixed cost is about $96,500. Bid costs are low for Granite Construction and high for firms with low participation rates, which matches the observed entry patterns.

The selection estimates suggest that observed play favors equilibria with cooperative or coordination-friendly features. Joint-profit maximizing equilibria are much more likely, dominated equilibria are much less likely, and high Nash-product pure equilibria are more likely. The pure-strategy coefficient is negative in the point estimate, so the model gives relatively more weight to mixed equilibria, though this coefficient is weaker than the joint-profit and dominance results.

7 Tables and figures

7.1 Table I: Example of a two-by-two game, and Table II: Bidder identities

Read Table I:

- Table I gives the generic two-by-two complete-information game used to explain the order-condition problem.
- One payoff for each player is normalized to zero.
- Even in this simplest case, there are more utility parameters than independent action probabilities when payoffs are unrestricted.

Read Table II:

- Table II reports the 10 largest CalTrans paving bidders by market share.
- Granite Construction is dominant: 27.2 percent share, 76 wins, and 244 bids entered.
- The four largest firms have shares of 27.2 percent, 10.4 percent, 6.6 percent, and 6.5 percent, motivating the focus on these firms in the entry game.

Economic message: Table I motivates why identification is difficult even in very small games. Table II shows why the empirical application can focus on a small strategic core of large bidders while treating smaller fringe bidders as exogenous.

TABLE I

EXAMPLE OF TWO-BY-TWO GAME

	L	R
T	$(0, 0)$	$(0, f_2(TR, x) + \epsilon_2(TR))$
B	$(f_1(BL, x) + \epsilon_1(BL), 0)$	$(f_1(BR, x) + \epsilon_1(BR), f_2(BR, x) + \epsilon_2(BR))$

Table I: Example of a two-by-two game.

TABLE II

BIDDER IDENTITIES AND SUMMARY STATISTICS

Company	Share	No. Wins	No. Bids Entered	Participation Rate	Total Bids for Contracts Awarded
Granite Construction Company	27.2%	76	244	58.9%	343,987,526
E. L. Yeager Construction Co Inc	10.4%	13	31	7.5%	132,790,460
Kiewit Pacific Co	6.6%	5	30	7.2%	112,057,627
M. C. M. Construction Inc	6.5%	2	6	1.4%	89,344,972
J. F. Shea Co Inc	3.3%	9	40	9.7%	43,030,861
Teichert Construction	3.3%	16	43	10.4%	40,177,076
W. Jaxon Baker Inc	2.9%	13	65	15.7%	37,702,631
All American Asphalt	2.2%	14	33	8.0%	30,764,962
Tullis And Heller Inc	2.1%	10	16	3.9%	27,809,535
Sully Miller Contracting Co	1.9%	17	49	11.8%	27,889,186

Table II: Bidder identities and summary statistics.

7.2 Tables III and IV: Bidding summary statistics and distance rankings

Read Table III:

- The average winning bid is about \$3.2 million.
- Winning bids are on average about 6.2 percent below the engineer's estimate.
- The average number of bidders is 4.68.
- The average normalized money left on the table is 6.79 percent of the engineer's estimate.

Read Table IV:

- The winning bidder is usually closer to the job site than lower-ranked bidders.
- The mean distance of the winning bidder is 47.47 miles, compared with 73.55 miles for the second-lowest bidder.
- Distance is therefore an economically meaningful cost shifter in this market.

Economic message: The bidding data show substantial asymmetric information and meaningful cost heterogeneity. Distance provides a key source of variation for both bid behavior and entry incentives.

TABLE III

BIDDING SUMMARY STATISTICS

	Obs	Mean	Std. Dev.	Min	Max
Winning bid	414	3,203,130	7,384,337	70,723	86,396,096
Markup: (winning bid – estimate)/estimate	414	−0.0617	0.1763	−0.6166	0.7851
Normalized bid: winning bid/estimate	414	0.9383	0.1763	0.3834	1.7851
Second lowest bid	414	3,394,646	7,793,310	84,572	92,395,000
Money on the table: second bid – first bid	414	191,516	477,578	68	5,998,904
Normalized money on the table: (second bid – first bid)/estimate	414	0.0679	0.0596	0.0002	0.3476
Number of bidders	414	4.68	2.30	2	19
Distance of the winning bidder	414	47.47	60.19	0.27	413.18
Travel time of the winning bidder	414	56.95	64.28	1.00	411.00
Utilization rate of the winning bidder	414	0.1206	0.1951	0.0000	0.9457
Distance of the second lowest bidder	414	73.55	100.38	0.19	679.14
Travel time of the second lowest bidder	414	82.51	97.51	1.00	614.00
Utilization rate of the second lowest bidder	414	0.1401	0.2337	0.000	0.9959

Table III: Bidding summary statistics.

TABLE IV

DISTANCE TO JOB SITE

	Mean	Std. Dev.	Min	Max
DIST1	47.47	60.19	0.27	413.18
DIST2	73.55	100.38	70.19	679.14
DIST3	75.47	95.56	0.13	594.16
DIST4	84.38	89.87	1.45	494.08
DIST5	76.12	86.33	1.25	513.31

Table IV: Distance to job site.

7.3 Tables V and VI: Bid function and logit entry model

Read Table V:

- The engineer's estimate is a powerful predictor of bids; the coefficient is close to one.
- Distance is positive and statistically significant in bid regressions.
- The fringe-firm dummy is positive, indicating that fringe firms submit substantially higher normalized bids.
- Project and firm fixed effects matter, consistent with project-level and firm-level heterogeneity.

Read Table VI:

- Entry probability falls with distance to the project.
- Firm effects are important: Granite has a much higher entry propensity than the baseline firm.
- Adding project fixed effects leaves distance strongly negative, supporting distance as an entry-cost shifter.

Economic message: Tables V and VI support the structural ingredients of the application. Distance shifts bidding costs and entry incentives, while firm effects capture persistent differences in participation costs and competitiveness.

TABLE V

Bid Function Regressions[a]

Variable	$b_{i,t}$	$b_{i,t}/\mathrm{EST}_t$	$b_{i,t}/\mathrm{EST}_t$	$b_{i,t}/\mathrm{EST}_t$
EST_t	1.025			
	(56.26)			
$\mathrm{DIST}_{i,t}$		0.000246	0.000223	
		(5.66)	(5.01)	
$\mathrm{UTIL}_{i,t}$			0.02539	
			(0.93)	
$\mathrm{FRINGE}_{i,t}$				0.4288
				(4.65)
Constant	−25,686	1.19	1.001	
	(0.56)	(94.9)	(79.98)	
Fixed effects	No	Project	Project	Project/firm
R^2	0.989	0.5245	0.5292	0.5321

[a]Number of observations is 1938; t-statistics are reported in parentheses. Fringe is a dummy variable that equals 1 for a fringe firm. Util denotes utilization rates.

Table V: Bid function regressions.

TABLE VI

Logit Model of Entry[a]

	I	II	III
Constant	−0.9067	−1.6811	
	(7.91)	(7.53)	
$\mathrm{DIST}_{i,t}$	−0.00218	−0.00322	−0.00854
	(5.42)	(5.66)	(4.85)
Granite		2.889	4.4537
		(13.28)	(7.31)
E. L. Yeager		—	—
Kiewit Pacific		−0.1527	1.1969
		(0.57)	(2.1)
M. C. M.		−1.786	−0.70779
		(3.94)	(1.12)
Fixed effects	No	No	Project
Observations	1656	1656	1068
Number of groups			261
Log-likelihood	−784.20	−511.86	−101.0728

[a]The dependent variable is whether one of the four largest firms in the industry decides to submit a bid in a particular procurement; t-statistics are reported in parentheses.

Table VI: Logit model of entry.

7.4 Tables VII and VIII: Markups, bid costs, and selection parameters

Read Table VII:

- The average estimated profit margin is 6.44 percent.
- The median margin is lower, 2.71 percent, indicating skewness in estimated margins.
- These magnitudes are consistent with observed construction-industry margins.

Read Table VIII:

- Joint-profit-maximizing equilibria receive a large positive selection coefficient.
- Pareto-dominated equilibria receive a large negative coefficient.
- The Nash-product coefficient is positive.
- The point estimate on pure strategy is negative, meaning mixed equilibria receive relatively more weight in the estimated selection rule.
- Estimated bid-preparation costs are lowest for Granite and highest for M. C. M. Construction.

Economic message: The final estimates show that the model is not only estimating entry costs; it is also learning which equilibria are selected in observed play. The pattern is consistent with tacit coordination on high-surplus, undominated outcomes.

TABLE VII

MARGIN ESTIMATES[a]

Variable	Num. Obs.	Mean	Std. Dev.	Median	25th Percentile	75th Percentile
Profit margin	1938	0.0644	0.1379	0.0271	0.0151	0.0520

[a]The markup is defined as 1 minus the ratio of the estimated cost, which is private information, to the actual bid.

Table VII: Margin estimates.

TABLE VIII

GAMES ESTIMATION RESULTS[a]

Variable	Mean	Median	Std. Dev.	95% Confidence Interval	
Equilibrium Selection Parameters (λ)					
Pure strategy	−1.3524	−1.5345	0.7979	−2.4903	0.1954
Joint profit maximizing	6.4365	6.4226	0.5321	5.6151	7.5149
Dominated	−5.3841	−5.3316	0.7002	−6.7164	−4.0986
Nash product	4.4143	4.2025	1.1017	2.9651	6.4836
Profit Scale					
Profit scale	0.0965	0.0954	0.0015	0.0954	0.0984
Bid Preparation Costs (θ_i)					
Granite Construction	0.2341	0.2393	0.0977	0.0679	0.4271
E. L. Yeager	1.4583	1.4757	0.0941	1.2563	1.6227
Kiewit Pacific	1.6751	1.6720	0.0511	1.5775	1.7789
M. C. M. Construction	2.4490	2.4360	0.1144	2.2547	2.6966

[a]Estimation and inference were performed using the LTE method of Chernozhukov and Hong (2003). A Markov chain was generated with 500 draws for each parameter. 409 importance games were used in the importance sampler for the 409 observations.

Table VIII: Games estimation results.

8 Short summary

- The paper studies static discrete games of complete information.
- Such games often have multiple equilibria, including mixed equilibria.
- The model therefore includes both payoff functions and an equilibrium selection mechanism.
- The selection mechanism assigns probabilities to equilibria based on equilibrium characteristics.
- The estimator computes all Nash equilibria and simulates action probabilities.
- Importance sampling makes the simulation practical by avoiding repeated equilibrium computation for every parameter value.
- MSM estimation uses differences between observed action indicators and simulated action probabilities as moments.
- Unrestricted nonparametric identification generally fails because there are too many payoff and selection parameters relative to action probabilities.
- Identification can be achieved with linear-index payoffs and rich covariate support, or with agent-specific payoff shifters.
- The application studies entry by large contractors into CalTrans highway procurement auctions.
- Bid data are used to recover markups and expected entry profits.
- The second stage estimates bid-preparation costs and the equilibrium selection mechanism.
- Observed equilibria are more likely to maximize joint profits, have high Nash product, avoid Pareto dominance, and, according to the point estimate, be mixed rather than pure.

9 Conclusion

9.1 Core conclusion

Bajari, Hong, and Ryan provide a framework for estimating complete-information discrete games when multiple equilibria and mixed strategies are empirically relevant. The key move is to treat equilibrium selection as a model primitive rather than an inconvenience.

9.2 Economic intuition

A complete-information game does not generally predict a unique outcome. If the same payoffs can support many equilibria, observed play reveals something about both incentives and coordination. The equilibrium selection mechanism captures this second component.

9.3 Methodological takeaway

The estimator is designed for realistic finite games. It computes all equilibria, handles mixed strategies, and uses importance sampling to reduce computation. This makes it possible to combine structural game theory with simulation-based econometrics.

9.4 Identification takeaway

The paper is clear about what cannot be learned without structure. Action frequencies alone do not identify arbitrary payoff functions and arbitrary selection rules. Identification requires normalizations plus either large-support linear-index restrictions or exclusion restrictions that shift one player's payoff independently.

9.5 Application takeaway

In California highway procurement auctions, entry behavior is consistent with meaningful bid-preparation costs and a nonrandom selection process. Equilibria that generate higher joint surplus and avoid domination are selected more often. This suggests that equilibrium selection itself can reveal economically meaningful coordination forces.

9.6 Limitations

The approach remains computationally intensive because it requires solving for all equilibria for many simulated utility draws. The empirical application therefore focuses on the four largest firms and treats fringe entry as predetermined. Identification also relies on strong assumptions about payoff shocks, support, and exclusion variation.

9.7 Takeaway

The enduring lesson is that empirical games need an explicit theory of equilibrium selection. Without it, the model may be unable to simulate outcomes or perform counterfactuals. With it, equilibrium multiplicity becomes a source of information rather than only a problem.